Module 7: Expanding Treatment & Design Structures

Generalized Block Designs

Limitiations of RCBD

- In an RCBD, each treatment appears *once* per block
  - limits ability to test treatment x block interaction
- Blocks reduce unexplained variability

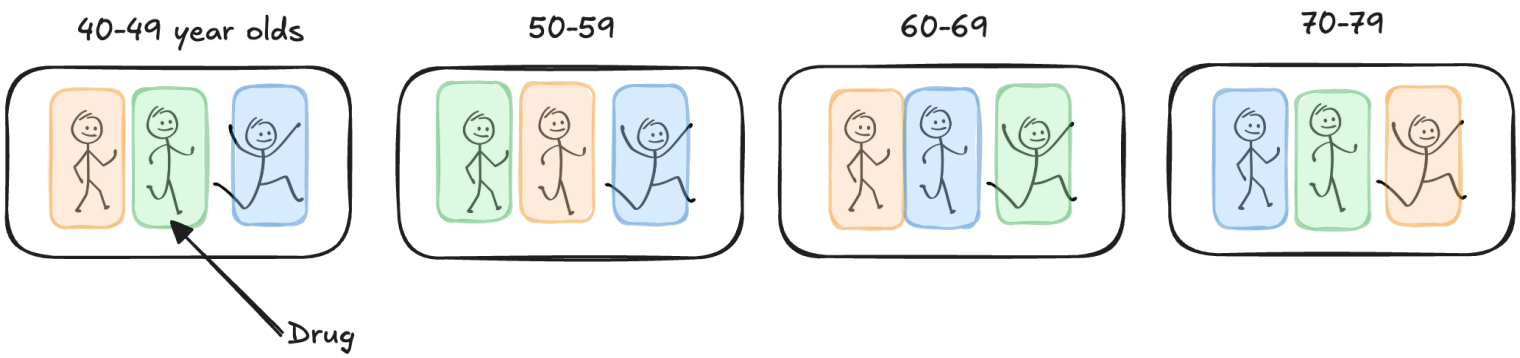
What if...

- One experimental unit per treatment x block isn't enough?
- We want to assess the treatment x block interaction?
- Blocks represent real populations (clinics, regions)?

Example 7.1: Cholesterol

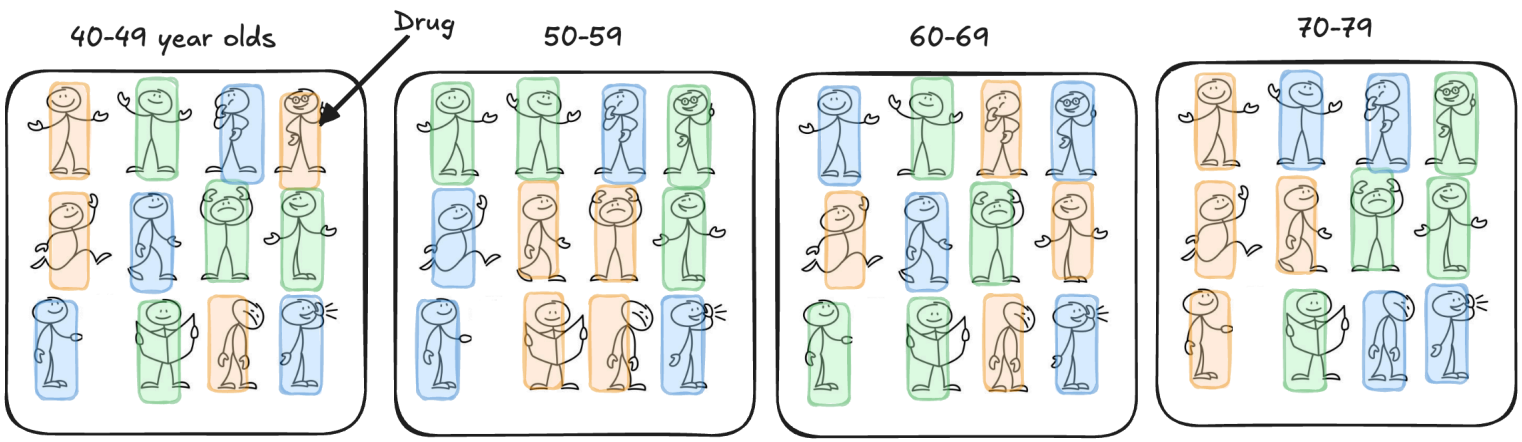
> Suppose we wish to compare three drugs (Placebo, Atorvastatin, Rosuvastatin) on their effectiveness at controlling cholesterol (change, post - pre) in adults. We decide to block on age group to reduce variability, using age ranges (40-49, 50-59, 60-69, 70-79).

As a RCBD, one adult in each age group is assigned each of the 3 drug treatments:



- Is one adult a good representation of all adults in that age range?
- Can we tell if the drug works differently for older vs younger patients?

Solution: Replicate *within* block!



Generalized Block Design

is used when we want or need to replicate within the block. This allows us to compare blocks while still accounting for block-to-block variation and assessing potential interactions between treatments and blocks.

Study Design

Treatment Structure One-way with three levels of drug (Placebo, Atorvastatin, Rosuvastatin); t = 3.

Experimental Structure Drug is randomly assigned to four adults (e.u.) within four age groups (40-49, 50-59, 60-69, 70-79) in a generalized block design for r = 16. The change in cholesterol (post - pre) is measured once for each adult (m.u.).

```r
cholesterol_data <- read.csv("data/07-cholesterol.csv")
head(cholesterol_data)
```

```
  Subject_ID Age_Group         Drug Rep_in_Cell Cholesterol_Change
1          1     40-49      Placebo           1          -3.752901
2          2     40-49      Placebo           2          -2.510216
3          3     40-49      Placebo           3          -4.624835
4          4     40-49      Placebo           4          -3.209782
5          5     40-49 Atorvastatin           1         -16.919306
6          6     40-49 Atorvastatin           2         -18.908985
```

Incorrect Analysis

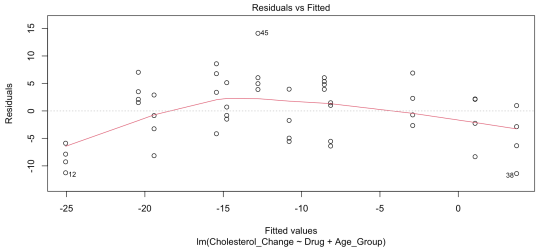
⚠️ Additive Model

$$y_{ij} = \mu + \tau_i + \rho_j + \epsilon_{ij}$$

```r
cholest_mod0 <- lm(Cholesterol_Change ~ Drug + Age_Group, data = cholesterol_data)
anova(cholest_mod0)
```

```
Analysis of Variance Table

Response: Cholesterol_Change
          Df Sum Sq Mean Sq F value    Pr(>F)
Drug       2 2319.0 1159.51  32.628 2.814e-09 ***
Age_Group  3 1025.7  341.90   9.621 5.879e-05 ***
Residuals 42 1492.5   35.54
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
plot(cholest_mod0, which = 1)
```



Include block x treatment interaction

$y_{ijk} = \mu + \tau_i + \rho_j + \tau\rho_{ij} + \epsilon_{ij}$ with ϵ_{ijk} iid $\sim N(0, \sigma^2)$

for $i = 1, 2, 3$, $j = 1, 2, 3, 4$, and $k = 1, 2, 3, 4$.

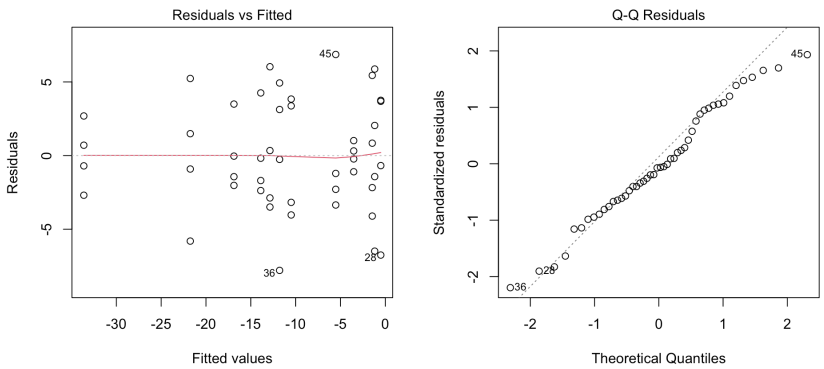
Source of Variation	DF

R: Include block x treatment interaction

```r
cholest_mod1 <- lm(Cholesterol_Change ~ Drug + Age_Group + Drug:Age_Group,
                   data = cholesterol_data)
anova(cholest_mod1)
```

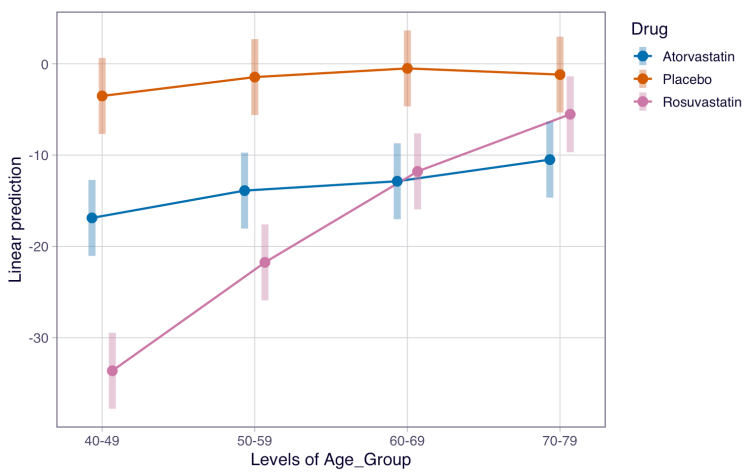
```r
par(mfrow = c(1,2))
plot(cholest_mod1, which = 1:2)
```

```
Analysis of Variance Table

Response: Cholesterol_Change
               Df  Sum Sq Mean Sq F value    Pr(>F)
Drug            2 2319.01 1159.51 68.9700 4.856e-13 ***
Age_Group       3 1025.70  341.90 20.3369 7.007e-08 ***
Drug:Age_Group  6  887.33  147.89  8.7967 6.338e-06 ***
Residuals      36  605.22   16.81
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```



R: Example 7.1 Recommendations

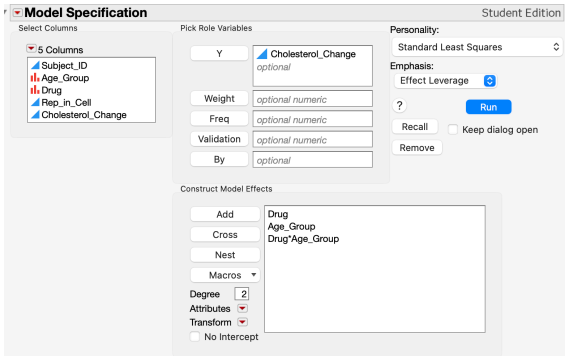
```r
library(emmeans)
library(multcomp)
emmip(cholest_mod1, Drug ~ Age_Group, CI = T)
```




```r
emmeans(cholest_mod1, ~ Drug | Age_Group) |>
  cld(Letters = LETTERS, decreasing = T, adjust = "tukey")
```

```
Age_Group = 40-49:
 Drug          emmean   SE df lower.CL upper.CL .group
 Placebo       -3.524 2.05 36    -8.66    1.609  A
 Atorvastatin -16.882 2.05 36   -22.02  -11.749   B
 Rosuvastatin -33.624 2.05 36   -38.76  -28.491    C

Age_Group = 50-59:
 Drug          emmean   SE df lower.CL upper.CL .group
 Placebo       -1.458 2.05 36    -6.59    3.676  A
 Atorvastatin -13.891 2.05 36   -19.02   -8.758   B
 Rosuvastatin -21.753 2.05 36   -26.89  -16.620    C

Age_Group = 60-69:
 Drug          emmean   SE df lower.CL upper.CL .group
 Placebo       -0.505 2.05 36    -5.64    4.628  A
 Rosuvastatin -11.790 2.05 36   -16.92   -6.657   B
 Atorvastatin -12.867 2.05 36   -18.00   -7.733   B

Age_Group = 70-79:
 Drug          emmean   SE df lower.CL upper.CL .group
 Placebo       -1.183 2.05 36    -6.32    3.950  A
 Rosuvastatin  -5.528 2.05 36   -10.66   -0.395  AB
 Atorvastatin -10.500 2.05 36   -15.63   -5.366   B

Confidence level used: 0.95
Conf-level adjustment: sidak method for 3 estimates
```

JMP: Include block x treatment interaction



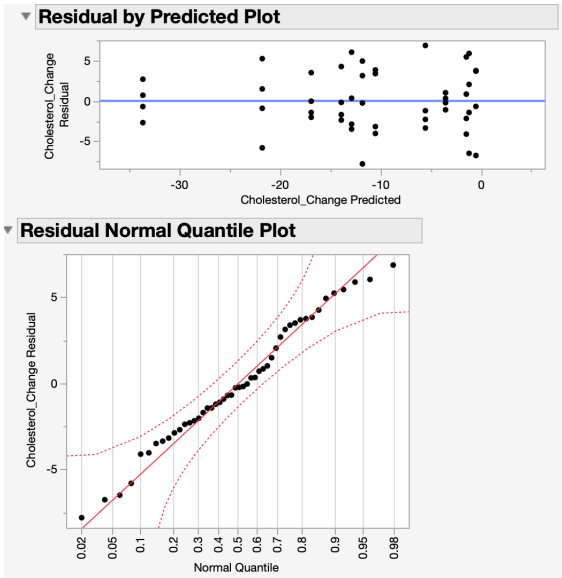
Analysis of Variance

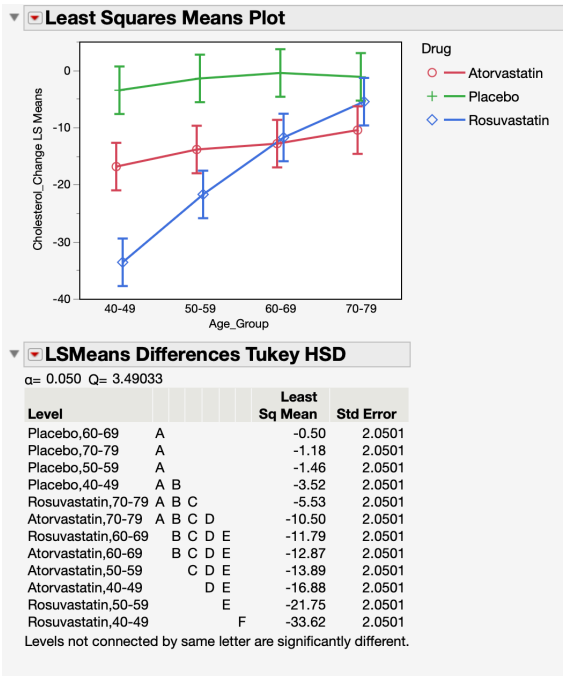
Source	DF	Sum of Squares	Mean Square	F Ratio
Model	11	4232.0337	384.730	22.8846
Error	36	605.2231	16.812	Prob > F
C. Total	47	4837.2568		<.0001*

Parameter Estimates

Effect Tests

Source	Nparm	DF	Sum of Squares	F Ratio	Prob > F
Drug	2	2	2319.0124	68.9700	<.0001*
Age_Group	3	3	1025.6961	20.3369	<.0001*
Drug*Age_Group	6	6	887.3253	8.7967	<.0001*





LSMeans Differences Tukey HSD

α= 0.050 Q= 3.49033

Level							Least Sq Mean	Std Error
Placebo,60-69	A						-0.50	2.0501
Placebo,70-79	A						-1.18	2.0501
Placebo,50-59	A						-1.46	2.0501
Placebo,40-49	A	B					-3.52	2.0501
Rosuvastatin,70-79	A	B	C				-5.53	2.0501
Atorvastatin,70-79	A	B	C	D			-10.50	2.0501
Rosuvastatin,60-69		B	C	D	E		-11.79	2.0501
Atorvastatin,60-69		B	C	D	E		-12.87	2.0501
Atorvastatin,50-59			C	D	E		-13.89	2.0501
Atorvastatin,40-49				D	E		-16.88	2.0501
Rosuvastatin,50-59					E		-21.75	2.0501
Rosuvastatin,40-49						F	-33.62	2.0501

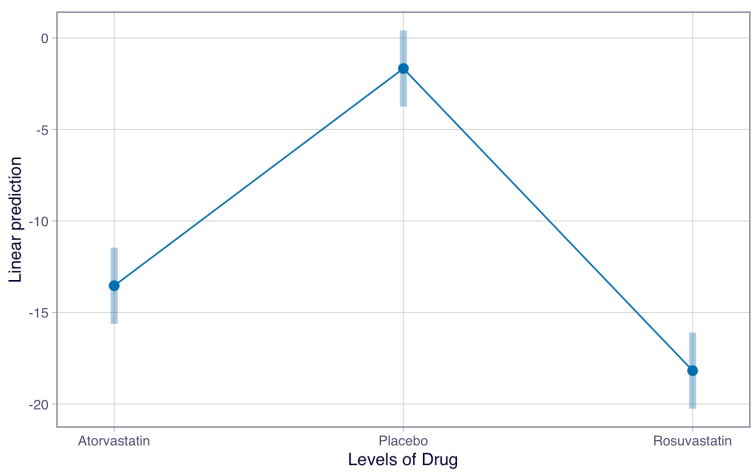
Levels not connected by same letter are significantly different.

Generalized Block Design vs Full Factorial

- Design: randomly assign treatments *within* blocks (kind of like a bunch of CRDs)
- Analysis: essentially the same as a full factorial ($\tau_i + \rho_j + \tau_{ij}$)

Misleading Main Effects (ignoring interaction)

```r
library(emmeans)
library(multcomp)
emmip(cholest_mod1, ~ Drug, CI = T)
```




```r
emmeans(cholest_mod1, ~ Drug) |>
  cld(Letters = LETTERS, decreasing = T, adjust = "tukey")
```

```
 Drug         emmean   SE df lower.CL upper.CL .group
 Placebo       -1.67 1.03 36    -4.23    0.899  A
 Atorvastatin -13.54 1.03 36   -16.10  -10.968   B
 Rosuvastatin -18.17 1.03 36   -20.74  -15.607    C

Results are averaged over the levels of: Age_Group
Confidence level used: 0.95
Conf-level adjustment: sidak method for 3 estimates
P value adjustment: tukey method for comparing a family of 3 estimates
significance level used: alpha = 0.05
NOTE: If two or more means share the same grouping symbol,
      then we cannot show them to be different.
      But we also did not show them to be the same.
```